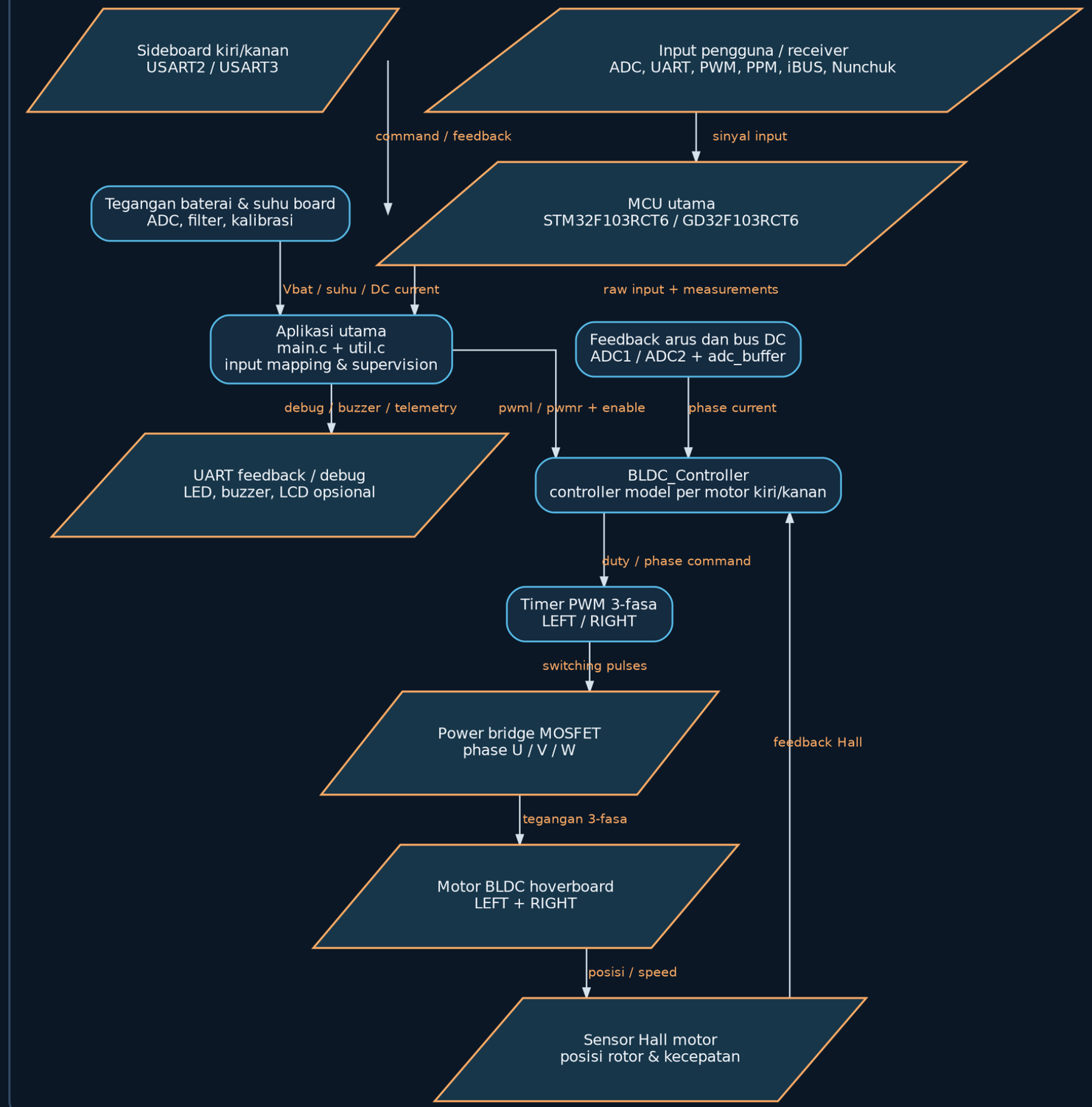
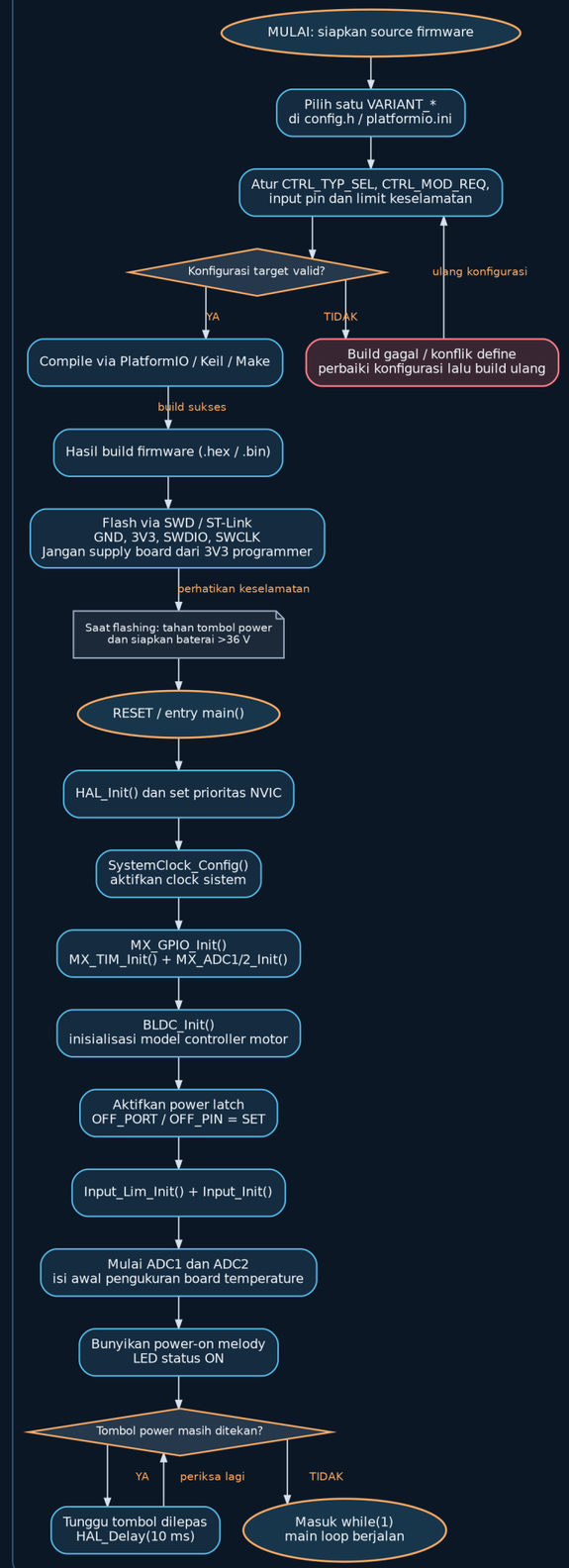
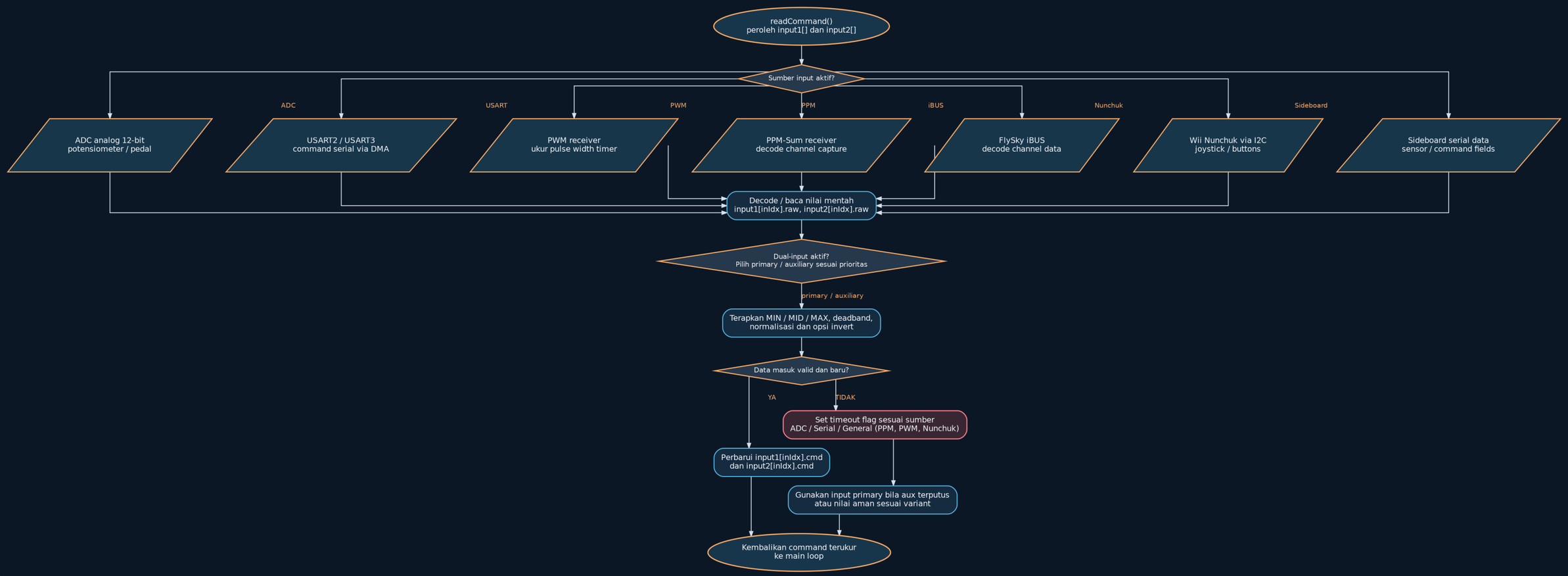
00 | ARSITEKTUR FIRMWARE HOVERBOARD FOC



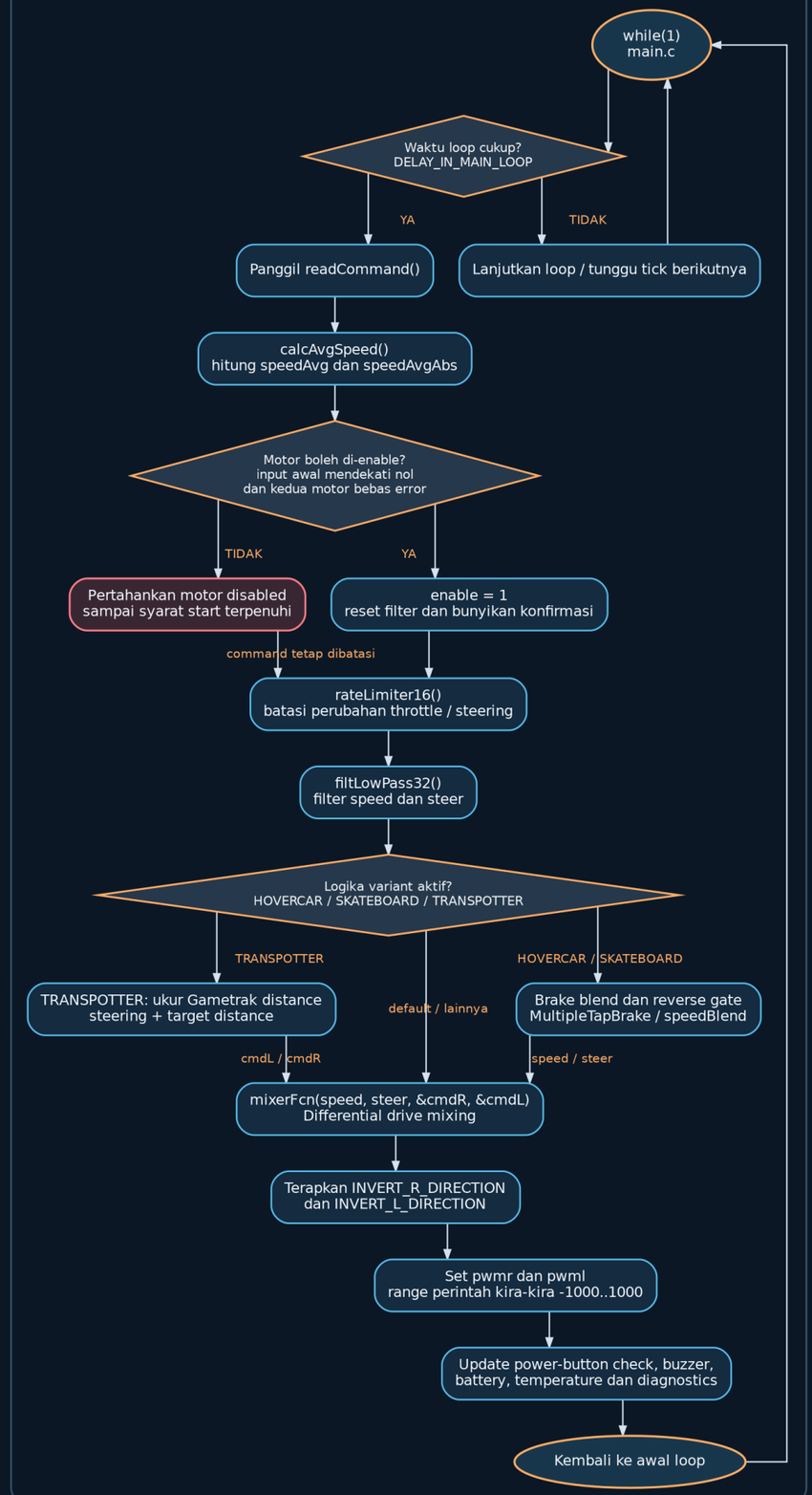
01 | BUILD, FLASH & INISIALISASI MCU



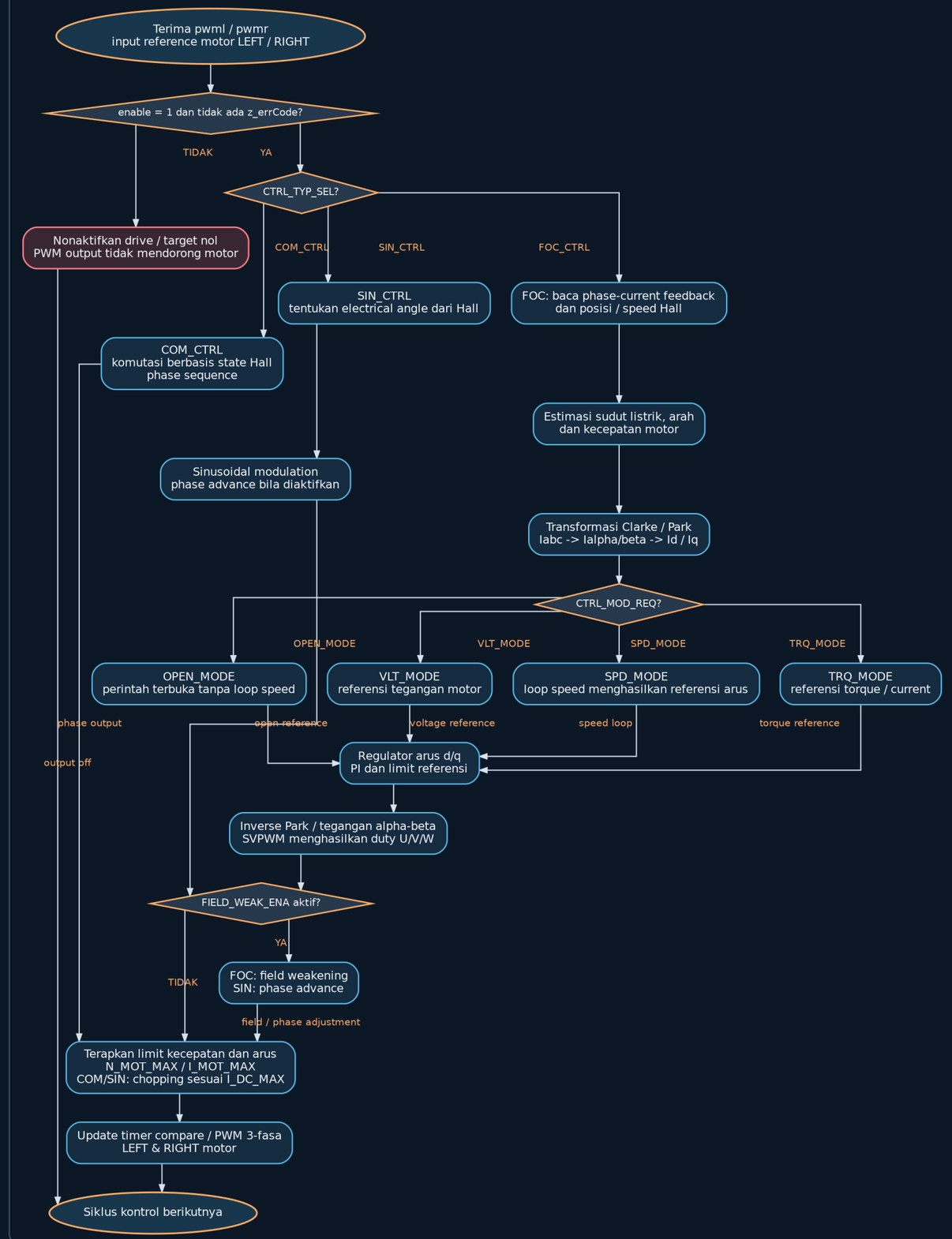
02 | AKUISISI INPUT, DECODE & INPUT VALIDATION



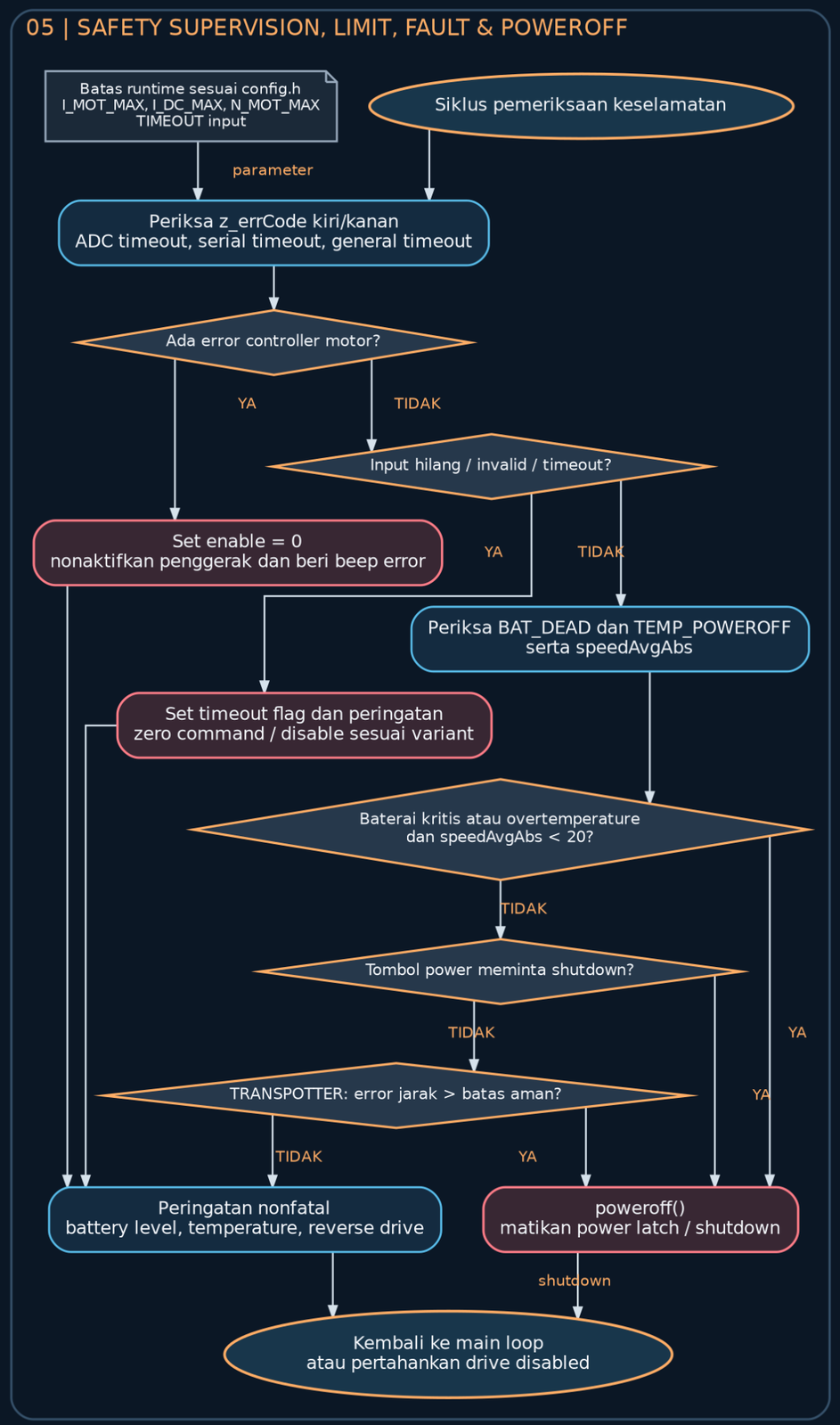
03 | MAIN LOOP, FILTER, MIXER & COMMAND MOTOR



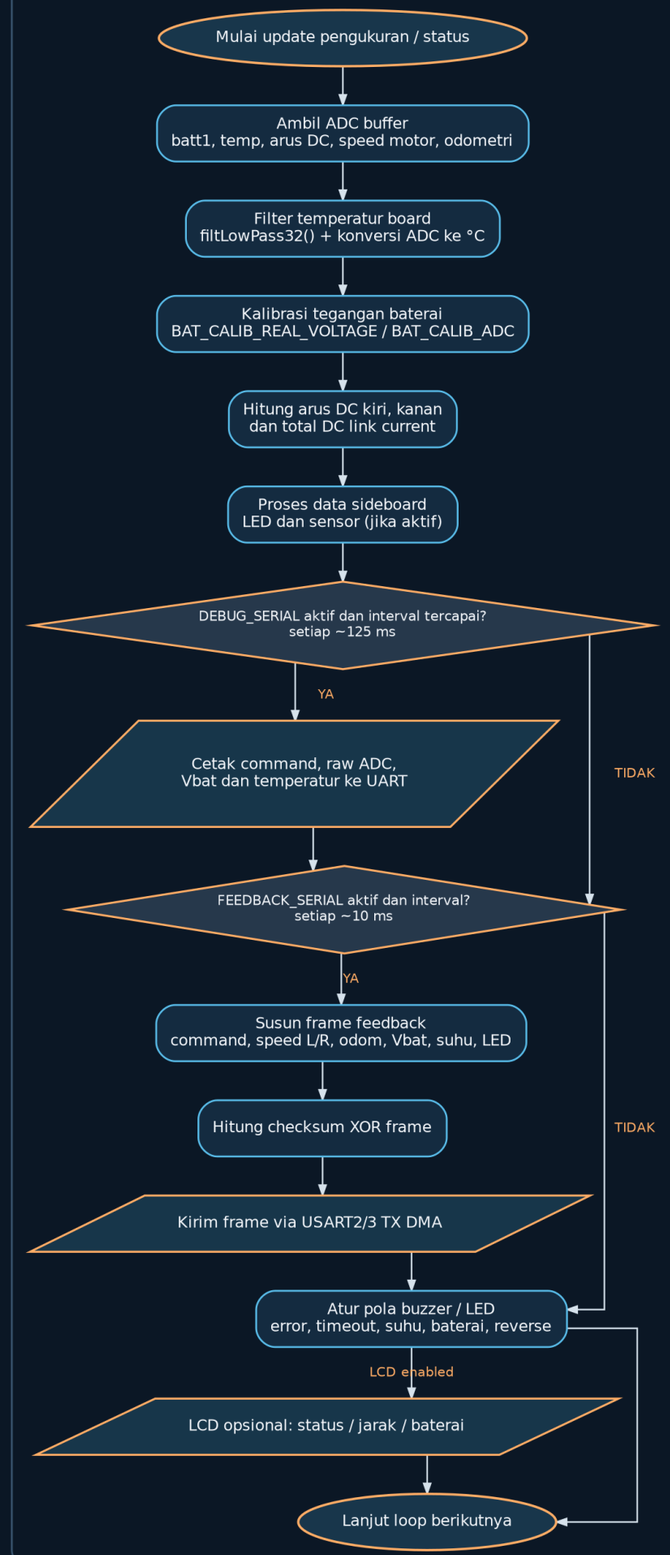
04 | DUAL BLDC CONTROL: COMMUTATION / SIN / FOC



05 | SAFETY SUPERVISION, LIMIT, FAULT & POWEROFF



06 | SENSOR, DIAGNOSTIK & SERIAL FEEDBACK



07 | VARIANT & PARAMETER DEFAULT (config.h)

I_MOT_MAX = 15 A; I_DC_MAX = 17 A
N_MOT_MAX = 1000 rpm
MOTOR_LEFT_ENA + MOTOR_RIGHT_ENA

PWM_FREQ = 16000 Hz; DEAD_TIME = 48
DELAY_IN_MAIN_LOOP = 5 ms (TRANSPOTTER: 2 ms)
TIMEOUT = 20

CTRL_MOD_REQ: OPEN / VOLTAGE / SPEED / TORQUE
Default: VLT_MODE

CTRL_TYP_SEL: COM_CTRL / SIN_CTRL / FOC_CTRL
Default: FOC_CTRL

VARIANT_ADC / USART / NUNCHUK / PPM / PWM / iBUS
HOVERCAR / HOVERBOARD / TRANSPOTTER / SKATEBOARD